Poultry nutrition evidence

Building the Future of African Livestock | Side Quests 11-14

Run 1395afb10eb726ef

Data: 200 complete monthly USDA ratios; 63 complete trial-performance pens, 9 per treatment. Incomplete ancillary endpoints were excluded explicitly.

One-month holdout MAE: selected ETS(AAA) 0.2401 versus naive 0.2333.

Three-month holdout MAE: selected seasonal ARIMA 0.4451 versus naive 0.4167.

These selected models did not improve the primary holdout metric. No statistical superiority claim is made from the small test window.

Primary FCR contrasts: none significant after six-contrast Holm adjustment. Non-significance is not equivalence.

Secondary ADG: LowCP-NoSBM minus CT = -4.519 g/bird/day, pointwise 95% CI [-7.089, -1.949], Holm p=0.005197.

This ADG finding remains significant in all 63 leave-one-pen-out analyses, under the same modeling assumptions.

Missing block identifiers prevent exact replication of the published blocked analysis. The estimand is a whole-diet effect.

13 internal control tests passed; independent Python numerical verification passed.

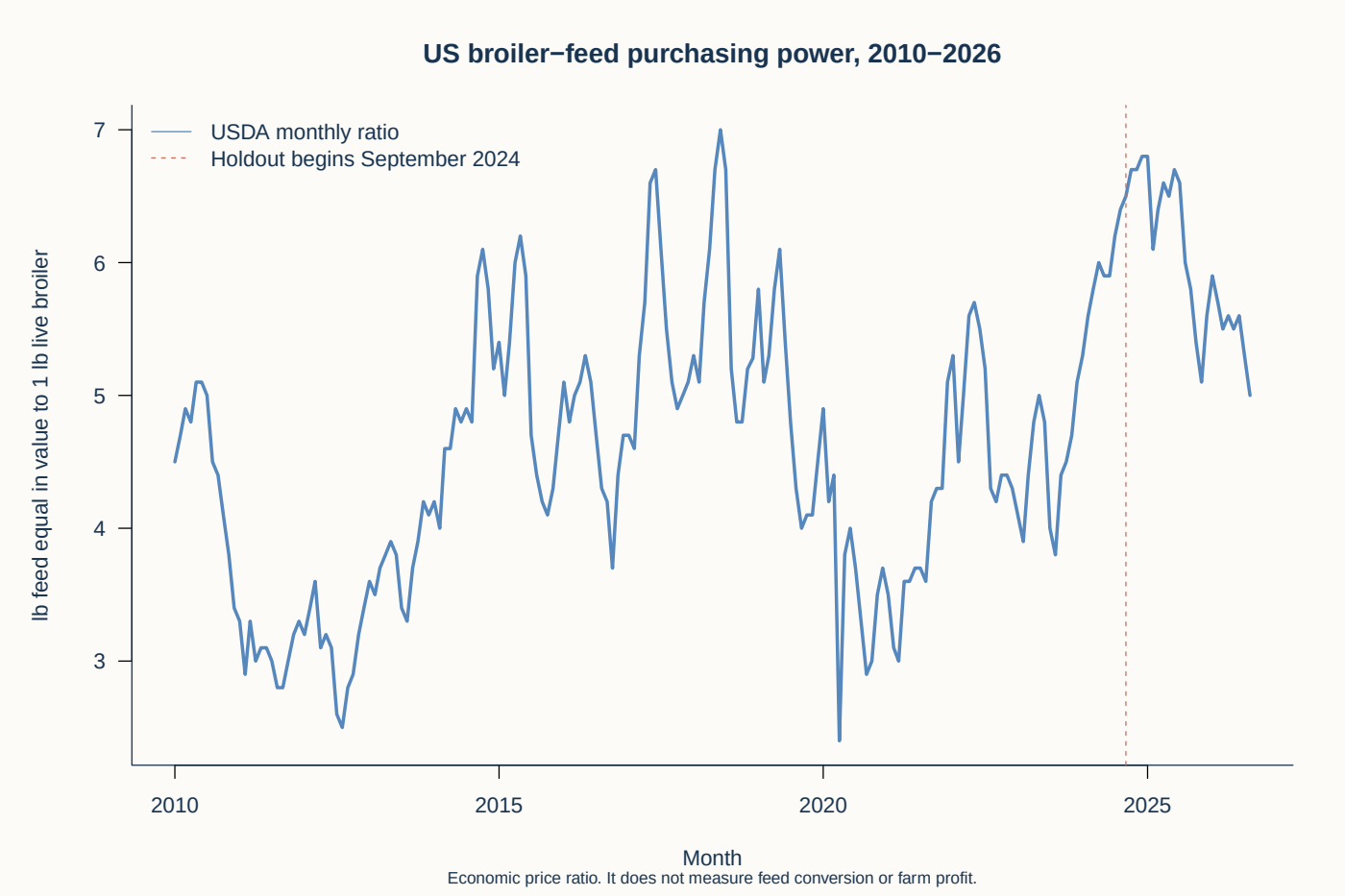
Source: https://www.ers.usda.gov/data-products/feed-grains-database/feed-grains-yearbook-tables

Article: https://doi.org/10.1016/j.animal.2026.101771

Data: https://doi.org/10.57745/1ZCQMN

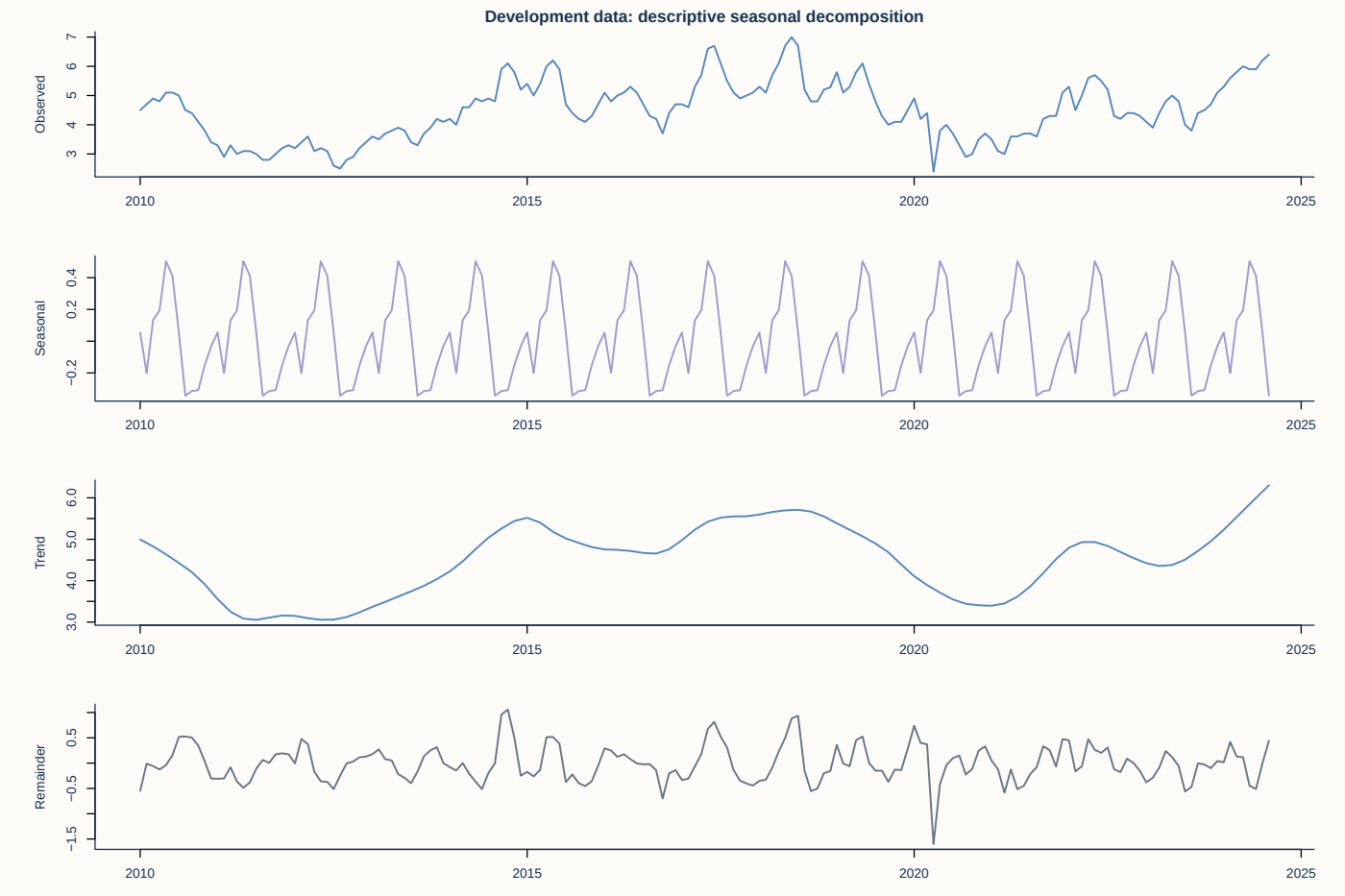
See the full HTML report for all charts, tables, interpretations and limitations. Governance documents distinguish implemented/tested controls from proposed deployment safeguards. No African treatment effect, autonomous operating approval or independent compliance certification is claimed.

Feed purchasing power



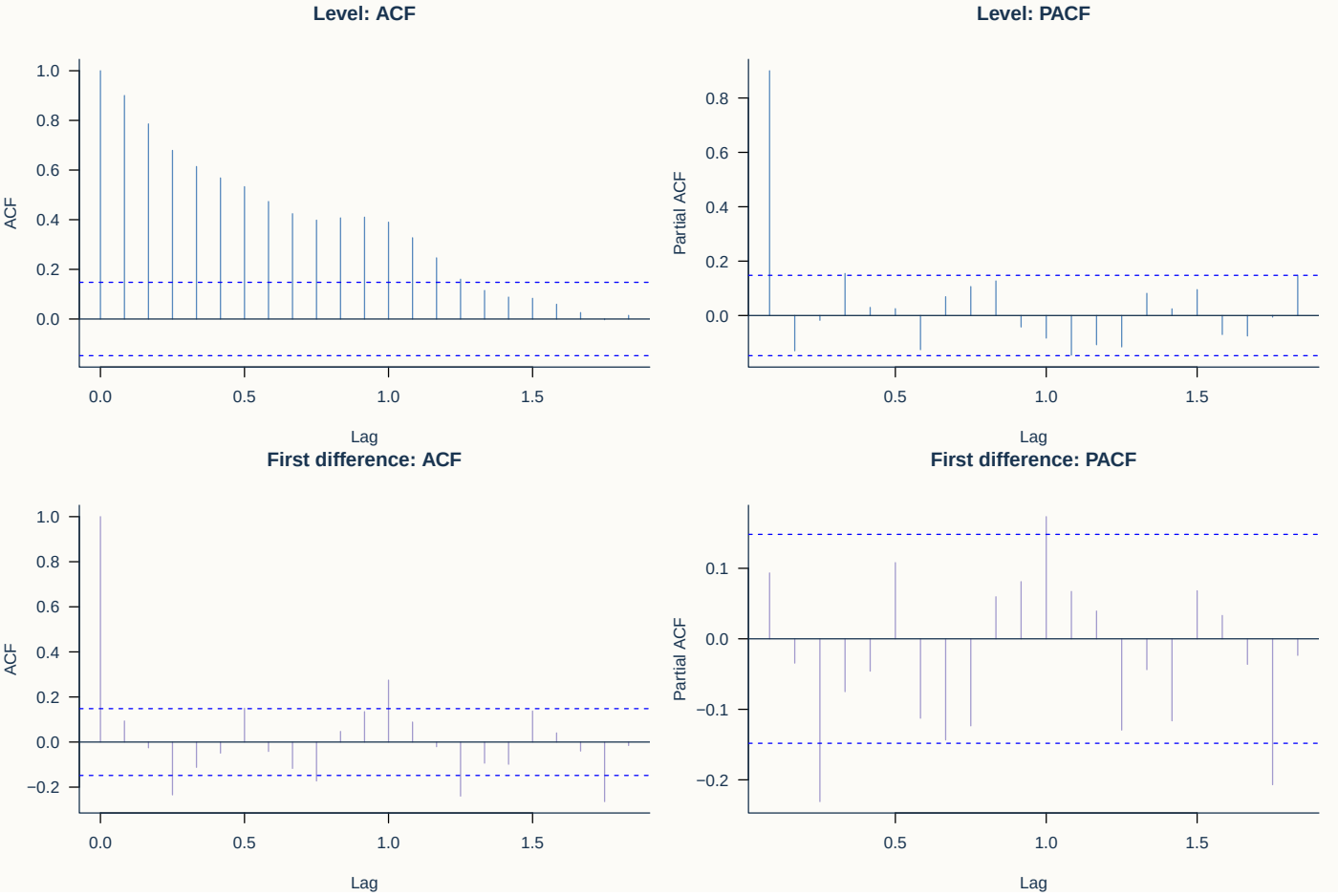
The ratio measures feed purchasing power, not feed cost per bird or biological feed conversion.

Seasonal decomposition



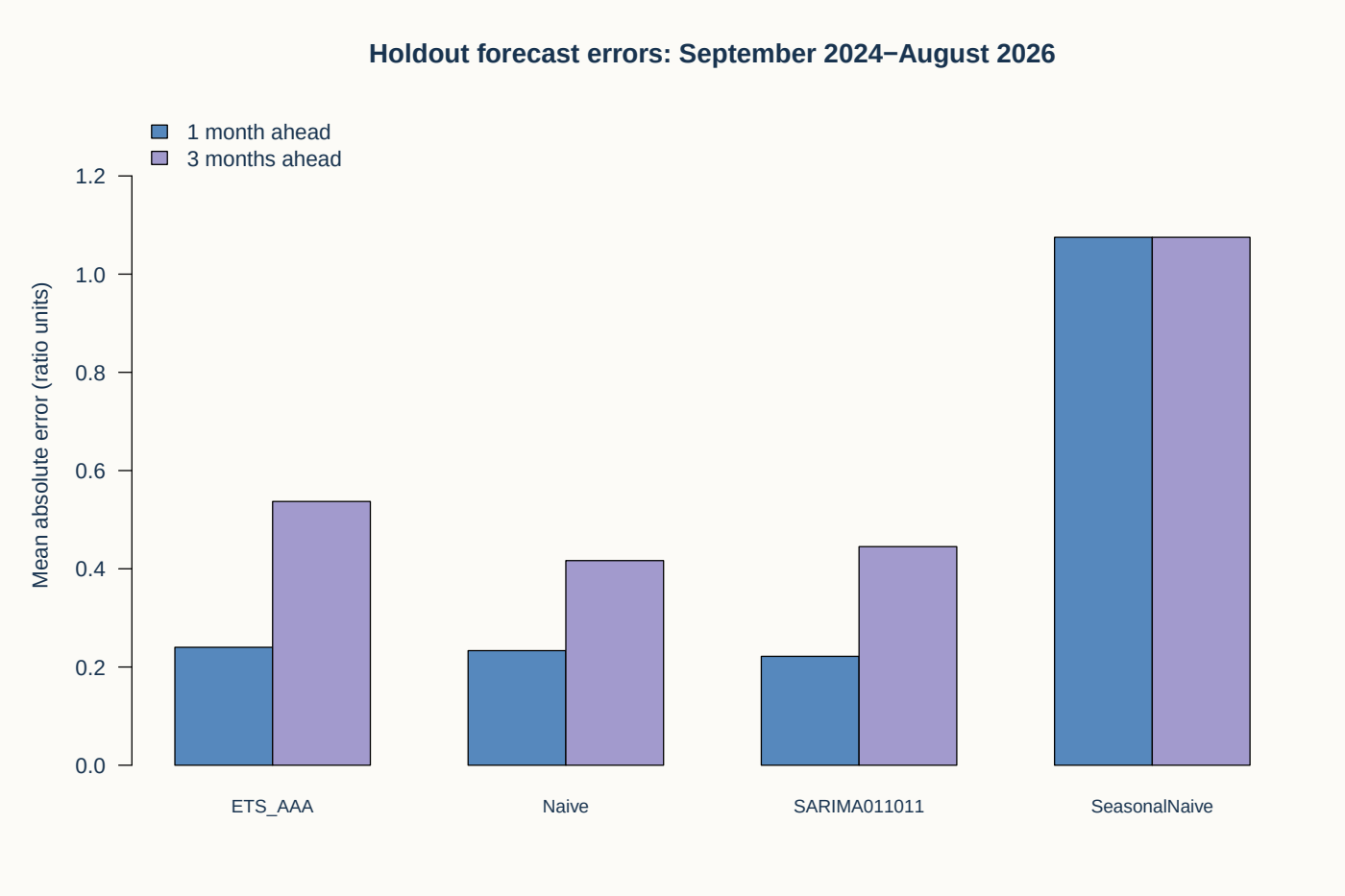
Decomposition describes historical structure; it does not establish causal effects.

Acf pacf



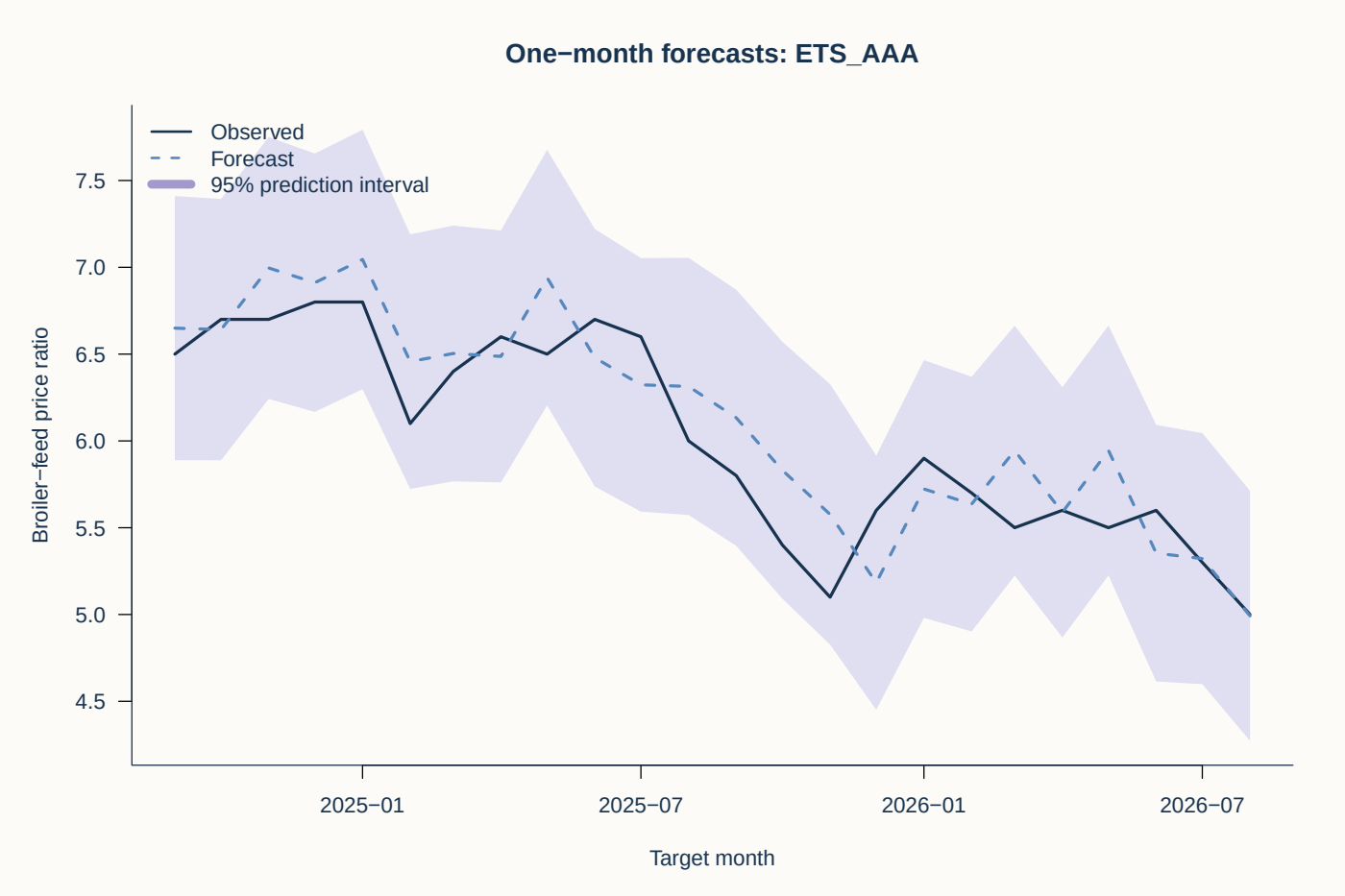
Autocorrelation diagnostics inform candidate specifications and differencing.

Holdout forecast accuracy



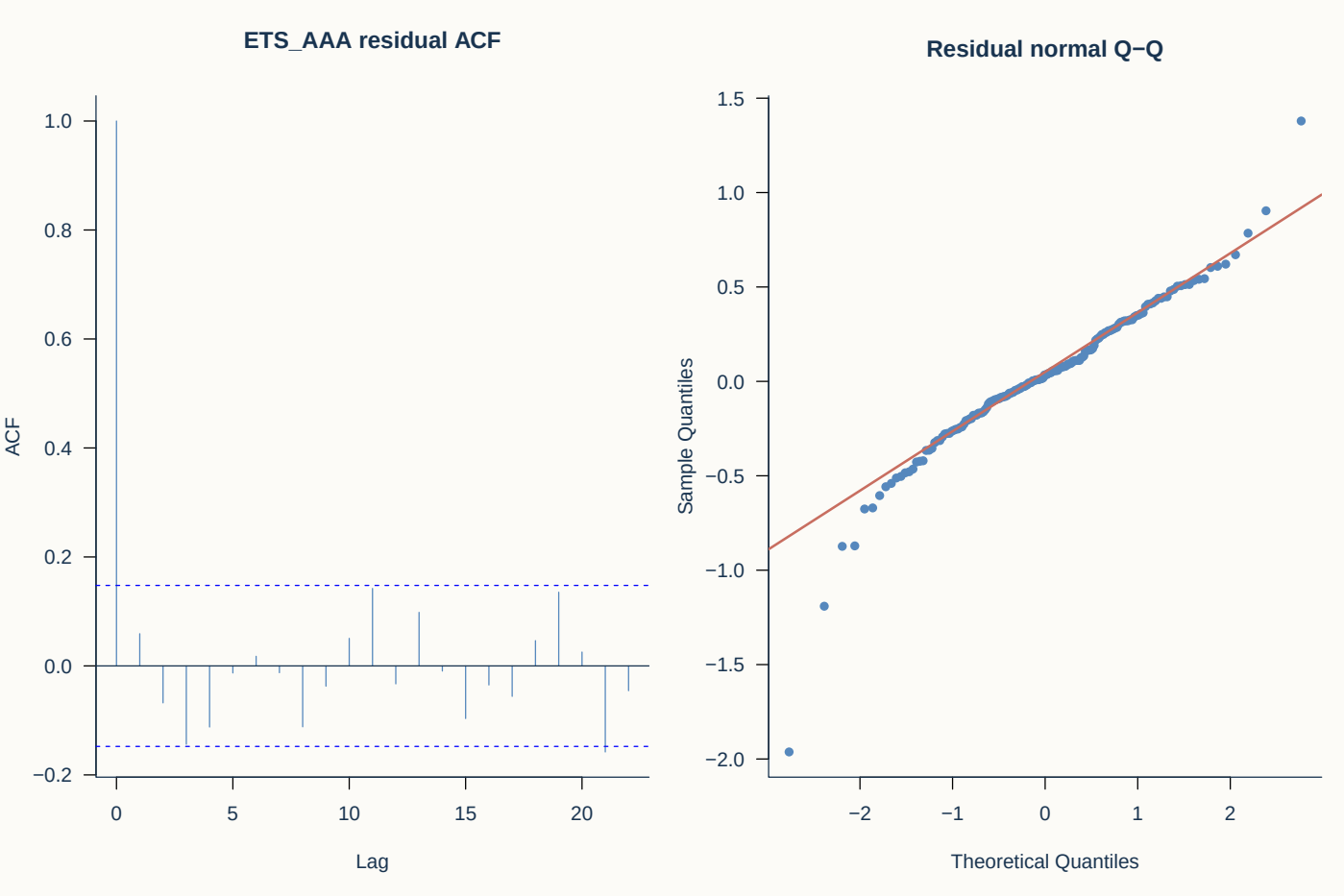
Selected models did not beat the naive baseline on held-out mean absolute error.

Holdout predictions



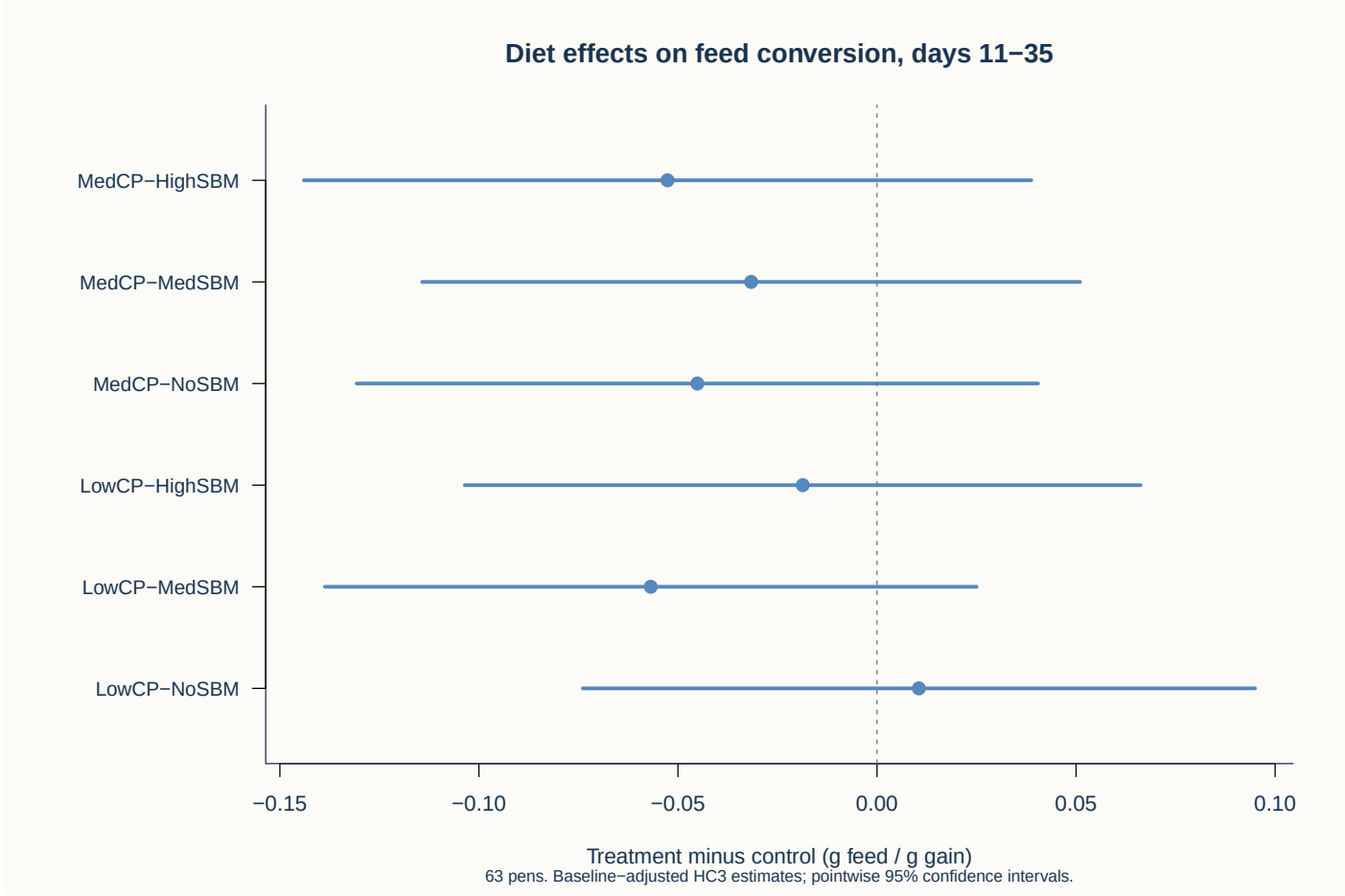
Target months were held out from model selection; forecasts use expanding histories.

Selected residual diagnostics



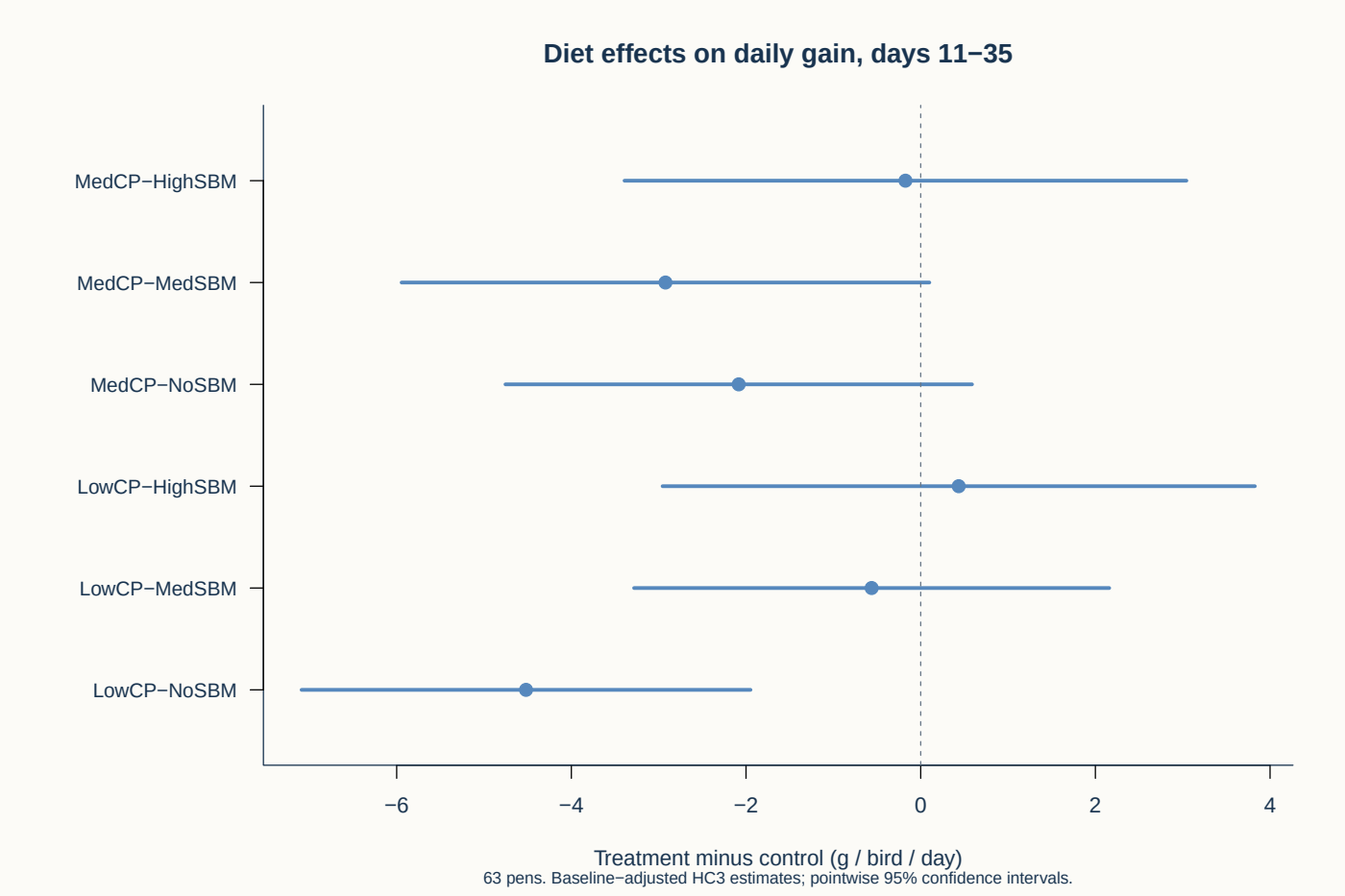
Residual dependence is a model limitation and must remain visible in interpretation.

Causal fcr



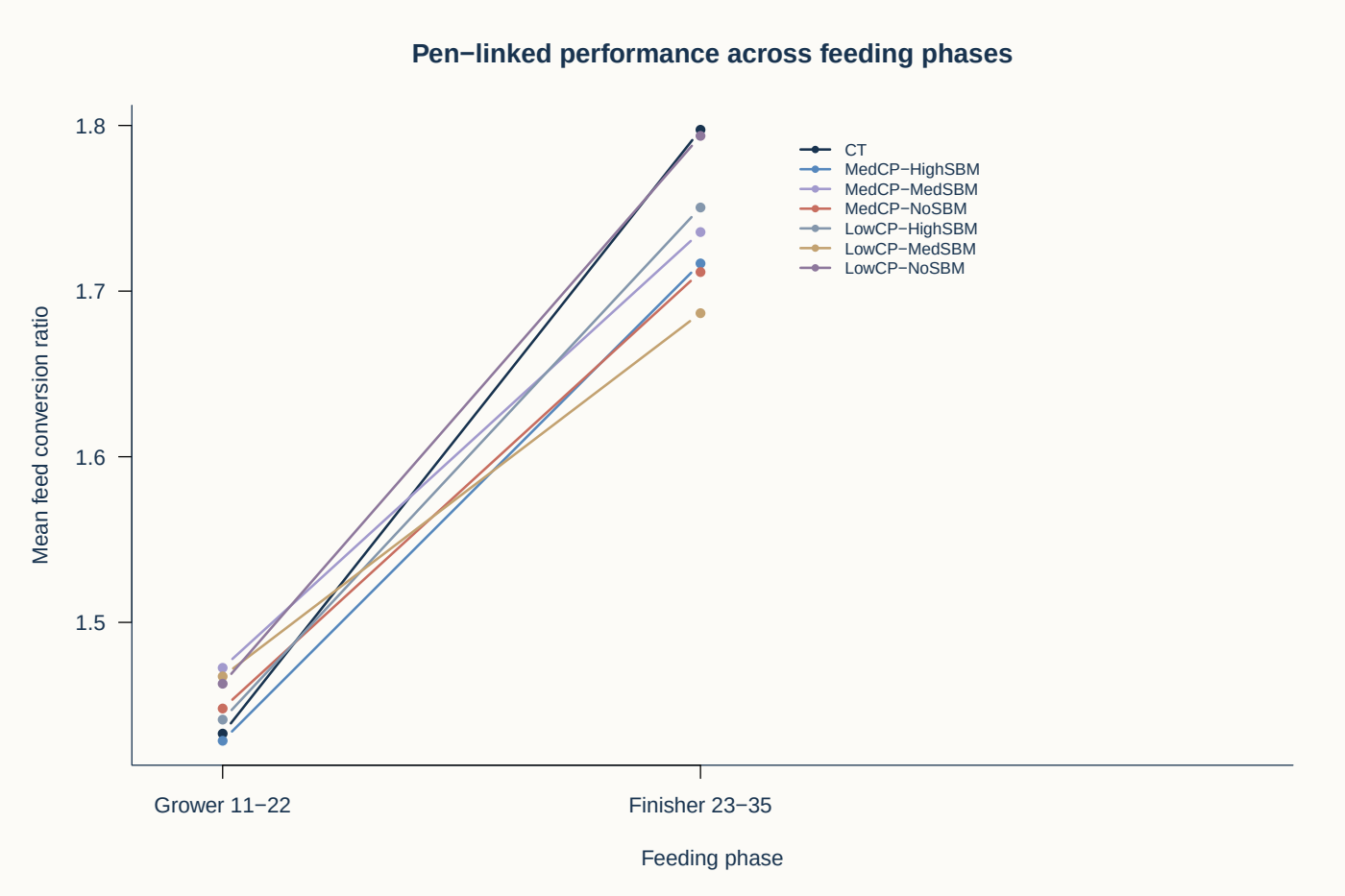
Whole-diet contrasts; no FCR contrast was significant after Holm correction. Non-significance is not equivalence.

Causal adg



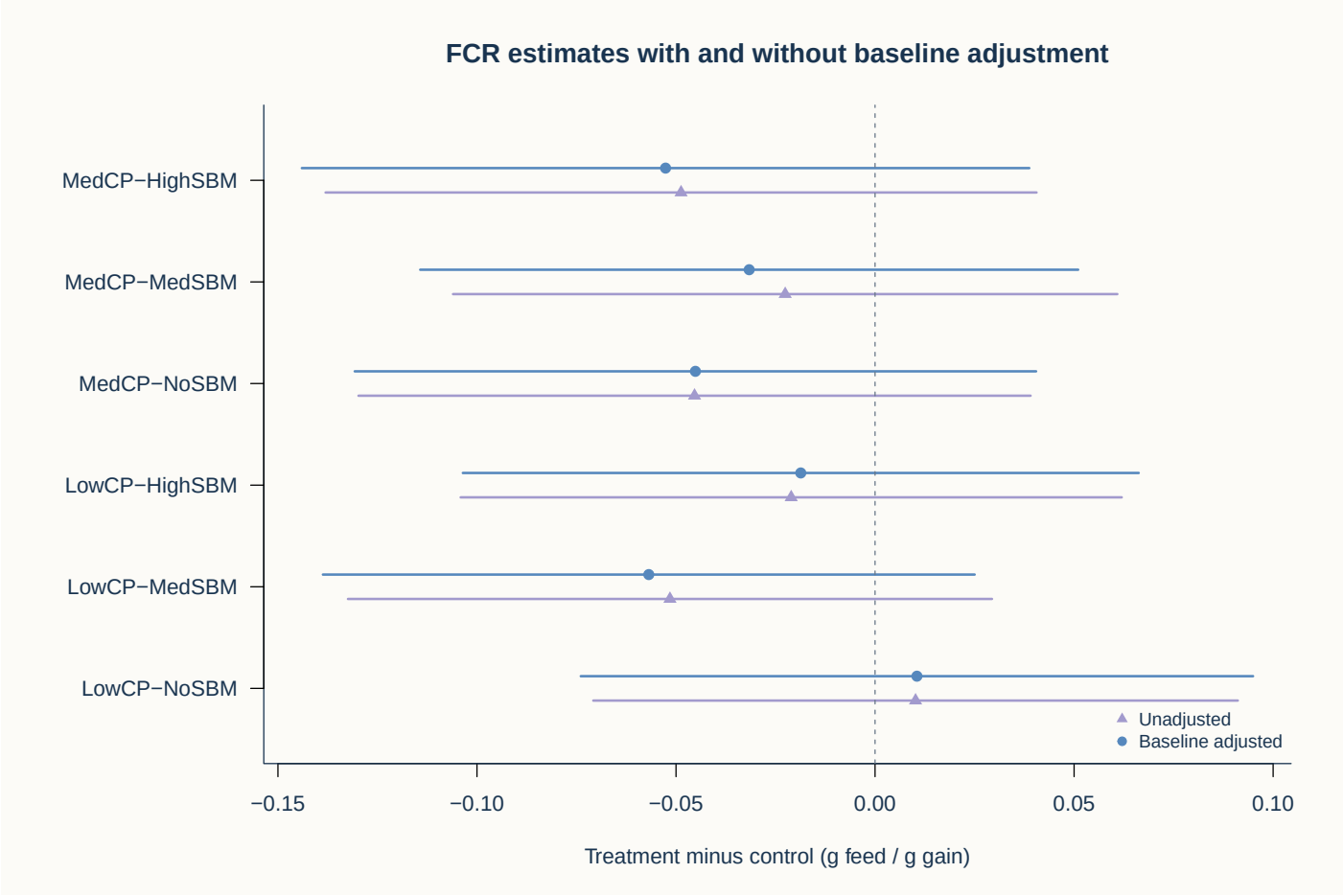
LowCP-NoSBM reduced daily gain relative to control under the stated analysis assumptions.

Phase performance



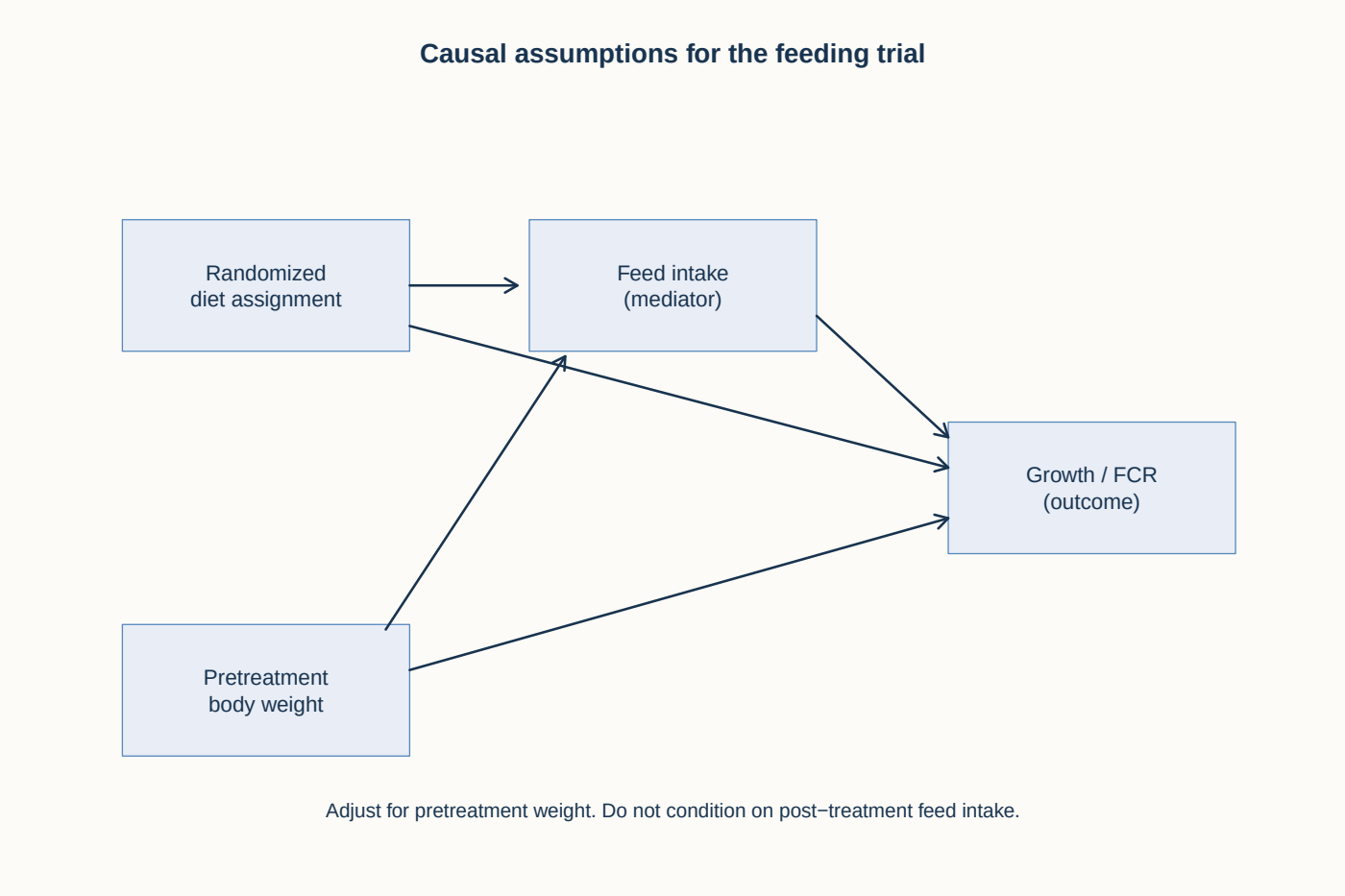
Two phases per pen: these measurements are longitudinal, not daily observations.

Sensitivity fcr



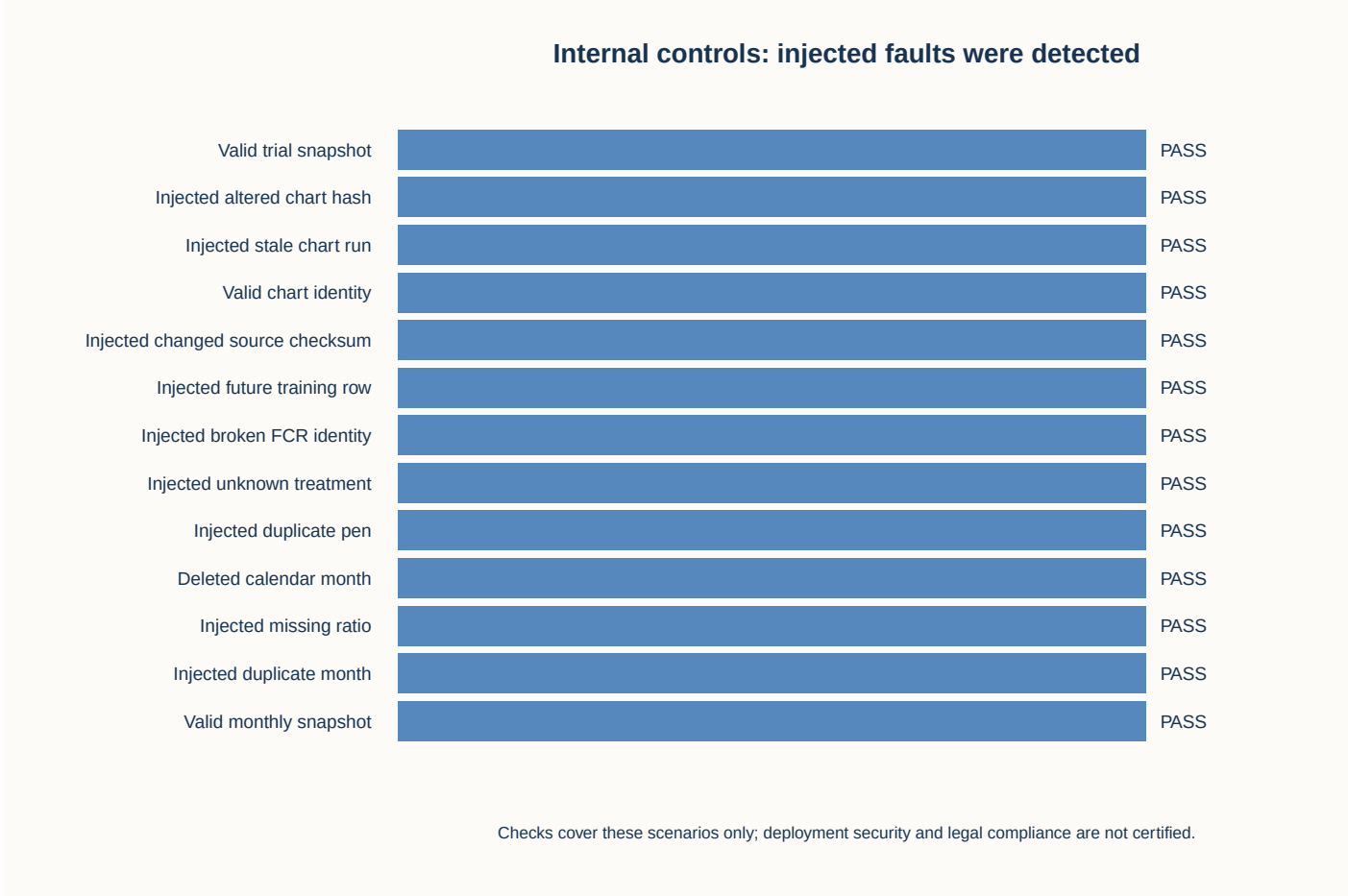
Sensitivity checks assess robustness but cannot recover missing randomized-block identifiers.

Trial dag



The causal graph documents assumptions; it does not by itself validate identification.

Control tests



These are internal controls for specified failures, not independent compliance certification.